

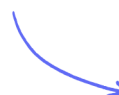
Structured Questions

Formulae & Equations

Formulae & Mass / Avogadro & the Mole / Full & Ionic Equations / Calculating Formulae

Medium (1 question)	/7
Hard (1 question)	/13
Total Marks	/20

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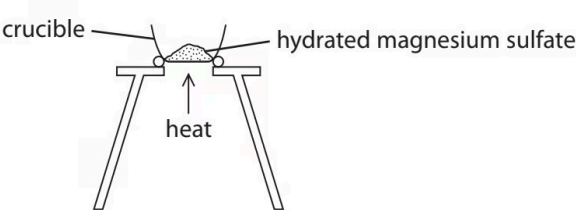


Medium Questions

- 1 (a)** This question is about the amount of water of crystallisation in hydrated magnesium sulfate, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

The value of x in the formula was determined in an experiment.

Procedure

Step 1	A crucible was weighed, a spatula measure of hydrated magnesium sulfate was added and the crucible was reweighed.
Step 2	The crucible containing the hydrated magnesium sulfate was heated using the apparatus shown. 
Step 3	After heating for two minutes, the crucible containing the magnesium sulfate was allowed to cool and was reweighed.

- i) Complete the table of results.

Measurement	Mass / g
Mass of empty crucible	21.21
Mass of crucible and hydrated magnesium sulfate before heating	26.71
Mass of crucible and magnesium sulfate after heating for two minutes	24.12
Mass of magnesium sulfate after heating for two minutes	
Mass of water lost	

(1)

ii) Use these results to calculate the value of x in $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

Give your answer to the nearest whole number. [A_r values: H = 1.0 O = 16.0 Mg = 24.3 S = 32.1]

(4)

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(5 marks)

(b) The correct value of x is greater than the value calculated in (a)(ii).

Suggest a way of improving the method to obtain a more accurate result, using the same apparatus. Justify your answer.

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(2 marks)

Hard Questions

1 (a) This question is about iron and some of its compounds.

Complete the table to show the numbers of subatomic particles in $^{56}\text{Fe}^{2+}$.

Number of protons	Number of neutrons	Number of electrons

(1 mark)

(b) A sample of iron contains the following isotopes.

Isotope	Percentage abundance
^{54}Fe	5.84
^{56}Fe	91.68
^{57}Fe	2.17
^{58}Fe	0.31

Calculate the relative atomic mass of this sample of iron. Give your answer to **three** significant figures.

(2 marks)

- (c) Magnesium reacts with aqueous iron(II) sulfate in a displacement reaction.

Write the **ionic** equation for this reaction. Include state symbols.

(2 marks)

- (d) 25.00 g of a compound contains 6.98 g of iron and 6.03 g of sulfur.

The remaining mass is oxygen.

Calculate the **empirical** formula of this compound.

[A_r values: O = 16.0 S = 32.1 Fe = 55.8]

(3 marks)

- (e) When 6.95 g of $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$ is heated, 2.00 g of iron(III) oxide, 0.80 g of sulfur dioxide and 1.00 g of sulfur trioxide are produced. The only other product is water.

Deduce the overall equation for the reaction using these data. State symbols are not required.

You **must** show your working.

[A_r values: H = 1.0 O = 16.0 S = 32.1 Fe = 55.8]

(5 marks)